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Substrate stripping method for semiconductor structure by irradiating AlGaN

Abstract

Provided is a method for stripping a substrate of a semiconductor structure, including: providing a substrate, a first AlN layer, a first AlGaN layer and a function layer from bottom to top; and irradiating the first AlGaN layer from the substrate with laser light to decompose the first AlGaN layer, such that the function layer is separated from the substrate and the first AlN layer.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a US National Phase of a PCT Application No. PCT/CN2020/120174 filed on Oct. 10, 2020, the entire contents of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(2) This present disclosure relates to the technical field of semiconductors and in particular, to methods for stripping a substrate of a semiconductor structure.

BACKGROUND

(3) Wide-bandgap semiconductor material, such as GaN-based material, as a typical representative of the third-generation semiconductor materials, has excellent properties of wide band gap, high pressure resistance, high temperature resistance, high electron saturation velocity, high drift velocity and easy formation of a high-quality heterostructure, which is suitable for manufacturing a high temperature, high frequency and high power electronic device.

(4) Due to lack of intrinsic substrates, a GaN-based device are generally manufactured on a heterogeneous substrate, such as a sapphire substrate, a silicon carbide substrate or a silicon substrate.

(5) In recent years, with the market demand for thinner and lighter devices, how to peel off a substrate of a GaN-based device and how to improve the peeling-off quality are technical problems that need to be solved urgently in the industry.

SUMMARY

(6) The object of the present disclosure is to provide a method for stripping a substrate of a semiconductor structure, which can improve the stripping quality and reduce the stripping cost.

(7) In order to achieve the above purpose, the present disclosure provides a method for stripping a

substrate of a semiconductor structure including: providing a substrate, a first AlN layer, a first AlGaIn layer and a function layer from bottom to top; and irradiating the first AlGaIn layer from the substrate with laser light to decompose the first AlGaIn layer, such that the function layer is separated from the substrate and the first AlN layer.

(8) In some embodiments, the method further includes providing a second AlN layer between the first AlGaIn layer and the function layer, where after the laser light irradiates the first AlGaIn layer from the substrate, the function layer and the second AlN layer are separated from the substrate and the first AlN layer.

(9) In some embodiments, the method further includes providing a second AlGaIn layer between the first AlGaIn layer and the function layer, where Al composition content in the second AlGaIn layer is higher than Al composition content in the first AlGaIn layer, and after the laser light irradiates the first AlGaIn layer from the substrate, the function layer and the second AlGaIn layer are separated from the substrate and the first AlN layer.

(10) It should be noted that the Al composition in the AlGaIn layer in the present disclosure refers to the percentage of the amount of Al element in the sum of the amount of Al element and the amount of Ga element, that is, the percentage of the number of Al atoms in the sum of the number of Al atoms and the number of Ga atoms.

(11) In some embodiments, Al component content in the first AlGaIn layer is less than 70%.

(12) In some embodiments, the first AlGaIn layer is a single-layer structure or a multi-layer structure.

(13) In some embodiments, the multi-layer structure includes first AlGaIn sublayers, and Al composition content in each of the first AlGaIn sublayers is different.

(14) In some embodiments, the multi-layer structure includes an AlGaIn/AlN alternating multi-layer superlattice structure.

(15) In some embodiments, a material of the substrate is sapphire.

(16) In some embodiments, the function layer includes: a light wave filtering structure or an LED structure.

(17) In some embodiments, the function layer includes: an LED structure, and a light emission wavelength of the LED structure is less than 350 nm.

(18) In some embodiments, the substrate includes a flat structure, and the first AlN layer includes a patterned structure: or both the substrate and the first AlN layer include a patterned structure.

(19) Compared with the prior art, the present disclosure has the following beneficial effects: 1) in the present disclosure, on the one hand, the first AlN layer and the first AlGaIn layer respectively correspond to a nucleation layer and a buffer layer when the function layer is epitaxially grown, to improve the quality of the function layer; and on the other hand, when the substrate is irradiated with laser light in certain bands, the first AlN layer is transparent to the laser light in the bands, but the first AlGaIn layer decomposes into nitrogen after absorbing the laser light in the bands, causing the first AlGaIn layer to be loose and porous and easy to separate, such that the substrate can be easily peeled off without damaging the function layer. 2) in some embodiments, there is a) a second AlN layer or b) a second AlGaIn layer between the first AlGaIn layer and the function layer, and the Al composition content in the second AlGaIn layer is greater than the Al composition content in the first AlGaIn layer. Compared with the solution b), the solution a) can improve the decomposition degree of the first AlGaIn layer, and the peeling effect is better. Compared with the solution a), the solution b) can improve the quality of the function layer. 3) in some embodiments, the first AlGaIn layer is a stacked structure, and the stacked structure includes multiple first AlGaIn sublayers, and the Al composition content in each of the first AlGaIn sub-layers is different. Compared with the solution in which the first AlGaIn layer has a single-layer structure and the composition content of Al is fixed, the advantage of multiple first AlGaIn sublayers is that the applicable laser wavelength range for peeling is large, or when the laser wavelength drifts, there is still a good peeling effect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a flowchart of a method for peeling off a substrate of a semiconductor structure according to a first embodiment of the present disclosure.
- (2) FIG. 2 and FIG. 3 are schematic diagrams of intermediate structures corresponding to processes in FIG. 1.
- (3) FIG. 4 is a schematic diagram of an intermediate structure corresponding to a method for peeling off a substrate of a semiconductor structure according to a second embodiment of the present disclosure.
- (4) FIG. 5 is a schematic diagram of an intermediate structure corresponding to a method for peeling off a substrate of a semiconductor structure according to a third embodiment of the present disclosure.
- (5) FIG. 6 is a schematic diagram of an intermediate structure corresponding to a method for peeling off a substrate of a semiconductor structure according to a fourth embodiment of the present disclosure.
- (6) To facilitate the understanding of the present disclosure, all reference signs present in the present disclosure are listed below: substrate **10** first AlN layer **11** first AlGaIn layer **12** reacted first AlGaIn layer **12'** function layer **13** first AlGaIn sublayer **121, 122** second AlGaIn layer **14** second AlN layer **15**

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (7) In order to make the above-mentioned objects, features and advantages of the present disclosure more obvious and understandable, embodiments of the present disclosure will be described in detail below with reference to the accompanying drawings.
- (8) FIG. 1 is a flowchart of a method for peeling off a substrate of a semiconductor structure according to a first embodiment of the present disclosure; and FIG. 2 and FIG. 3 are schematic diagrams of intermediate structures corresponding to processes in FIG. 1.
- (9) First, referring to step S1 in FIG. 1 and as shown in FIG. 2, a substrate **10**, a first AlN layer **11**, a first AlGaIn layer **12** and a function layer **13** are provided from bottom to top.
- (10) A material of the substrate **10** may be a material such as sapphire, silicon carbide, silicon or diamond, preferably a material with high transmittance to laser light used in the subsequent step S2.
- (11) A material of the function layer **13** may be a group III nitride-based material, such as at least one of GaN, AlGaIn, InGaIn, and AlInGaIn. The function layer **13** may correspond to a heterojunction structure in the HMET (High Electron Mobility Transistor) device, that is, the function layer **13** may include: a barrier layer and a buffer layer: or the function layer **13** may correspond to an LED structure in an LED device, that is, the function layer **13** may include: a P-type semiconductor layer, an N-type semiconductor layer and a single quantum well layer/a multiple quantum well layer/quantum dots/quantum wires between the P-type semiconductor layer and the N-type semiconductor layer.
- (12) In an embodiment, a light emission wavelength of the LED structure may be less than 350 nm.
- (13) In another embodiment, the LED structure can function with a pair of Bragg mirrors as a light wave filtering structure.
- (14) It should be noted that, in the embodiments, a material is represented by chemical elements, but a molar ratio of each chemical element in the material is not limited. For example, the GaN material contains Ga element and N element, but the molar ratio of Ga element and N element is not limited.
- (15) The first AlN layer **11** can serve as a nucleation layer when epitaxially growing the function layer **13**. The nucleation layer can alleviate the problems of lattice mismatch and thermal mismatch

between the epitaxially grown function layer **13** and the substrate **10**.

(16) The first AlGaIn layer **12** can serve as a buffer layer when epitaxially growing the function layer **13**. The buffer layer can reduce the dislocation density and defect density of the epitaxially grown function layer **13** and improve the crystal quality.

(17) In this embodiment, the first AlGaIn layer **12** includes a single-layer structure, and a material of the single-layer structure may be represented as AlGaIn.

(18) Next, referring to step S2 in FIG. 1 and as shown in FIG. 2 and FIG. 3, the first AlGaIn layer **12** is irradiated from the substrate **10** with laser light to decompose the first AlGaIn layer **12**, such that the function layer **13** is separated from the substrate **10** and the first AlN layer **11**.

(19) When the substrate **10** is irradiated by laser light in certain wavelength bands, the first AlN layer **11** is transparent to the laser light in the wavelength bands, but the first AlGaIn layer **12** will decompose into nitrogen gas after absorbing the laser light in the wavelength bands, and the reacted first AlGaIn layer **12'** is loose, porous and easy to be separated, such that the substrate **10** can be easily peeled off without damaging the function layer **13**.

(20) Research shows that, in the first AlGaIn layer **12**, the Al composition content is less than 70%, that is, when the percentage of the content of Al element in the first AlGaIn layer **12** to the sum of the contents of Al element and Ga element is less than 70%, correspondingly the wavelength range of the laser light with better decomposition degree is from 200 nm to 300 nm. Further, when the percentage of the content of Al composition in the first AlGaIn layer **12** is less than 40%, the wavelength range of the laser light being from 250 nm to 280 nm has better the decomposition degree.

(21) Sapphire has a high transmittance in the above-mentioned 200 nm-300 nm wavelength band, and can be used as a preferred material for the substrate **10**.

(22) It should be noted that, in this embodiment, the range includes endpoint values.

(23) FIG. 4 is a schematic diagram of an intermediate structure corresponding to a method for peeling off a substrate of a semiconductor structure according to a second embodiment of the present disclosure.

(24) Referring to FIG. 4, the method for peeling off the substrate according to the second embodiment is substantially the same as the method for peeling off the substrate according to the first embodiment, the difference is that the first AlGaIn layer **12** is a multi-layer structure, the multi-layer structure includes multiple first AlGaIn sublayers **121** and **122**, and the Al composition content in each of the first AlGaIn sublayers **121** and **122** is different.

(25) Compared with the solution in which the first AlGaIn layer **12** is a single-layer structure and the Al composition content is fixed, the advantage of the solution for multi-layer structure is that the applicable laser light wavelength range for peeling is large, or when the laser light wavelength drifts, there is still a good peeling effect. In addition, the multi-layer structure of the first AlGaIn layer can also protect the upper function layer **13**.

(26) In some embodiments, the multi-layer structure may further include: AlGaIn/AlN alternating multi-layer superlattice structure. In the superlattice structure, the Al composition content in each of the AlGaIn layers is different or the same.

(27) FIG. 5 is a schematic diagram of an intermediate structure corresponding to a method for peeling off a substrate of a semiconductor structure according to a third embodiment of the present disclosure.

(28) Referring to FIG. 5, the method for peeling off a substrate according to the third embodiment is substantially the same as the method for peeling off a substrate according to the first and second embodiments, and the difference is that a second AlGaIn layer **14** is provided between the first AlGaIn layer **12** and the function layer **13**, where the Al composition content in the second AlGaIn layer **14** is higher than the Al composition content in the first AlGaIn layer **12**; and after the laser light irradiates the first AlGaIn layer **12** from the substrate **10**, the function layer **13** and the second AlGaIn layer **14** are separated from the substrate **10** and the first AlN layer **11**.

(29) The laser light can pass through the first AlN layer **11** and cannot pass through the first AlGaN layer **12**. In some embodiments, the laser light can pass through both the first AlN layer **11** and the second AlGaN layer **14**, and cannot pass through the first AlGaN layer **12**, such that the damage of the second AlGaN layer **14** by the laser light can be avoided.

(30) The higher the Al content in the AlGaN layer, the better the quality of the function layer **13**, so the solution of this embodiment can improve the quality of the function layer **13**.

(31) FIG. **6** is a schematic diagram of an intermediate structure corresponding to a method for peeling off a substrate of a semiconductor structure according to a fourth embodiment of the present disclosure.

(32) Referring to FIG. **6**, the method for peeling off a substrate according to the fourth embodiment is substantially the same as the method for peeling off a substrate according to the third embodiment, and the difference is that a second AlN layer **15** is provided between the first AlGaN layer **12** and the function layer **13**; and after the laser light irradiates the first AlGaN layer **12** from the substrate **10**, the function layer **13** and the second AlN layer **15** are separated from the substrate **10** and the first AlN layer **11**. In other words, the second AlN layer **15** replaces the second AlGaN layer **14**.

(33) Compared with the solution in the third embodiment, in this solution the thickness of the AlGaN layer is reduced, so under the same laser power, the decomposition degree of the first AlGaN layer **12** can be improved, and the peeling effect is better.

(34) In some embodiments, the methods for peeling off a substrate according to embodiments 1 to 4 can also be applied to a patterned substrate.

(35) For example, in an embodiment, the substrate **10** includes a flat structure, and the first AlN layer **11** includes a patterned structure.

(36) In another embodiment, both the substrate **10** and the first AlN layer **11** include a patterned structure. For example, the substrate **10** and the first AlN layer **11** are patterned in the same process; or the substrate **10** is patterned first, the first AlN layer **11** is epitaxially grown on the patterned substrate **10**.

(37) Although the present disclosure discloses the above contents, the present disclosure is not limited thereto. One of ordinary skill in the art can make various variants and modifications to the present disclosure without departing from the spirit and scope of the present disclosure. Therefore, the protection scope of the present disclosure should be set forth by the appended claims.

Claims

1. A method for stripping a substrate of a semiconductor structure, comprising: providing a substrate, a first AlN layer, a first AlGaN layer and a function layer from bottom to top; and irradiating the first AlGaN layer from the substrate with laser light to decompose the first AlGaN layer, such that the function layer is separated from the substrate and the first AlN layer, wherein the substrate comprises a flat structure, and the first AlN layer comprises a patterned structure; wherein the first AlGaN layer is a multi-layer structure; wherein the multi-layer structure comprises an AlGaN/AlN alternating multi-layer superlattice structure, wherein in the superlattice structure, Al composition content in each of AlGaN layers is different.
2. The method according to claim 1, further comprising: providing a second AlN layer between the first AlGaN layer and the function layer, wherein after the laser light irradiates the first AlGaN layer from the substrate, the function layer and the second AlN layer are separated from the substrate and the first AlN layer.
3. The method according to claim 1, further comprising: providing a second AlGaN layer between the first AlGaN layer and the function layer, wherein Al composition content in the second AlGaN layer is higher than Al composition content in the first AlGaN layer, and after the laser light irradiates the first AlGaN layer from the substrate, the function layer and the second AlGaN layer

are separated from the substrate and the first AlN layer.

4. The method according to claim 1, wherein Al component content in the first AlGaIn layer is less than 70%.

5. The method according to claim 1, wherein a material of the substrate is sapphire.

6. The method according to claim 1, wherein the function layer comprises: a light wave filtering structure or an LED structure.

7. The method according to claim 1, wherein the function layer comprises: an LED structure, and a light emission wavelength of the LED structure is less than 350 nm.
